

Technical Paradigms of Agricultural Electricity Distribution

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Invited talk

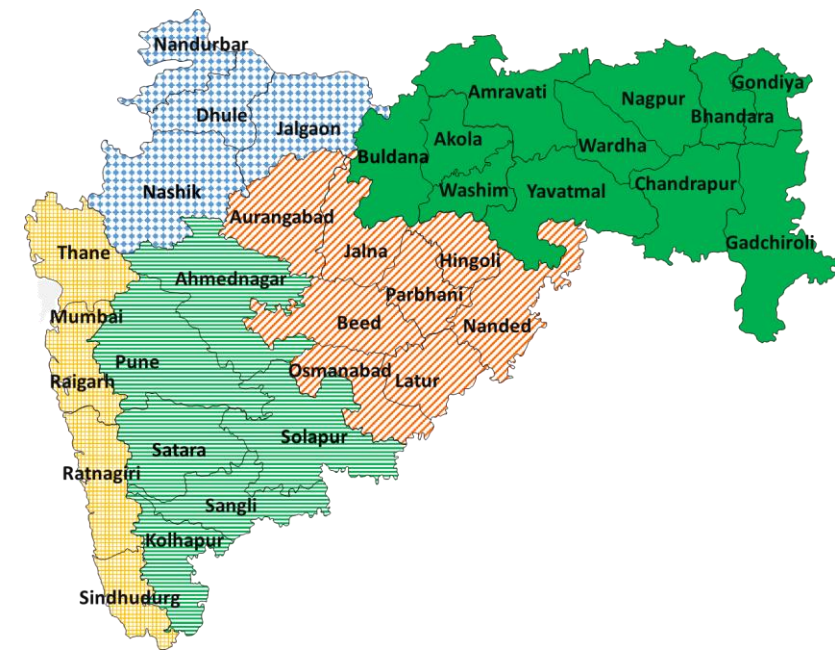
Maharashtra Electricity Regulatory Commission

August 31, 2020



Rationale for this work

- Research in agricultural distribution systems
- Power systems problems from the field which are appropriate for the context
- Maharashtra has 43 lakh grid connected farmers; 1 – 1.5 lakh new connections every year
- Quality of supply issues for farmers, heavily loaded systems;
- Delays in getting connections on Ag feeders over several years is common
- HVDS, Offgrid solar PV pumps, Solar Ag feeders - Many new paradigms being experimented
- A rationale analysis of the various systems, and solutions to improve the current system



A techno-economic analysis of various paradigms, as well as solutions within the current system

Outline of this presentation

1. Analysis of different modes of electricity supply for irrigation

- Agricultural feeders, HVDS, Solar Ag feeders, Offgrid solar PV pumps, Netmetered solar PV pumps; A techno-economic analysis considering field conditions in the state

From a published paper (Details in next slide)

2. Investigations and solutions for improvements within current Ag feeder systems

Work being done by IITB as part of an MoU with Project on Climate Resilient Agriculture (PoCRA) or 'नानाजी देशमुख कृषि संजीवनी प्रकल्प' (a GoM / World Bank project)

Overview of part (1)

- HVDS, Offgrid solar PV pumps, Solar Ag feeders, Net-metered solar pumps - Many new paradigms being experimented in the field
- Analysis and comparison of different technologies. Comparison should include : **Techno-economic, environmental** and **socio-political considerations**
- Invited talk in MERC on August 31, 2020
- A high level analysis based on agro-climatic regions of Maharashtra



A synopsis of the publication: Jadhav, Priya Jayawant, Namita Sawant, and Athira Madhusudhana Panicker. "Technical paradigms in electricity supply for irrigation pumps: Case of Maharashtra, India." *Energy for Sustainable Development* 58 (2020): 50-62. <https://doi.org/10.1016/j.esd.2020.07.005>

First step - financial comparison of all systems

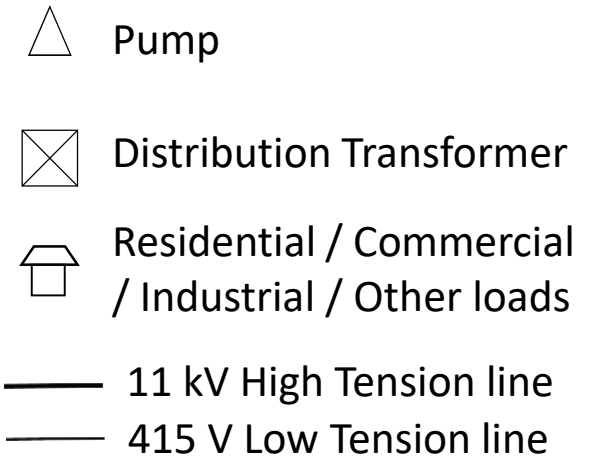
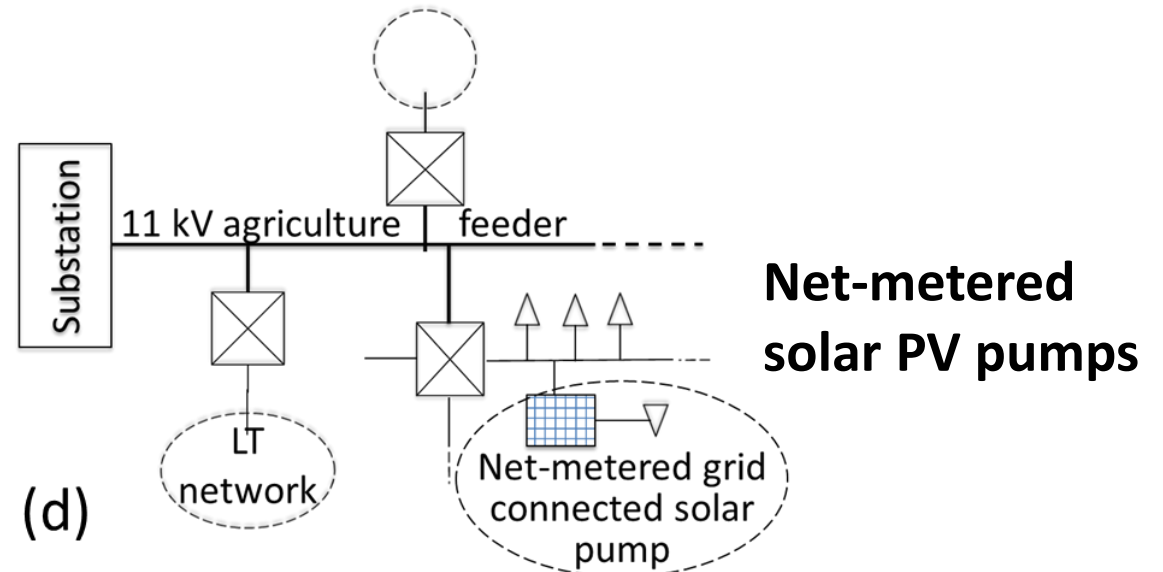
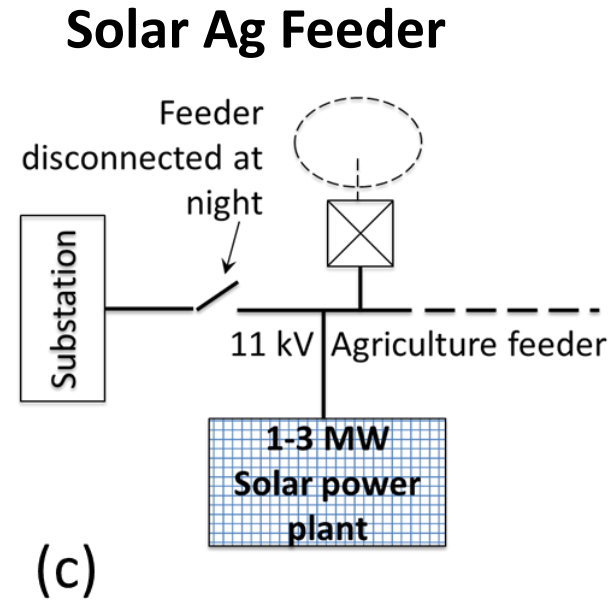
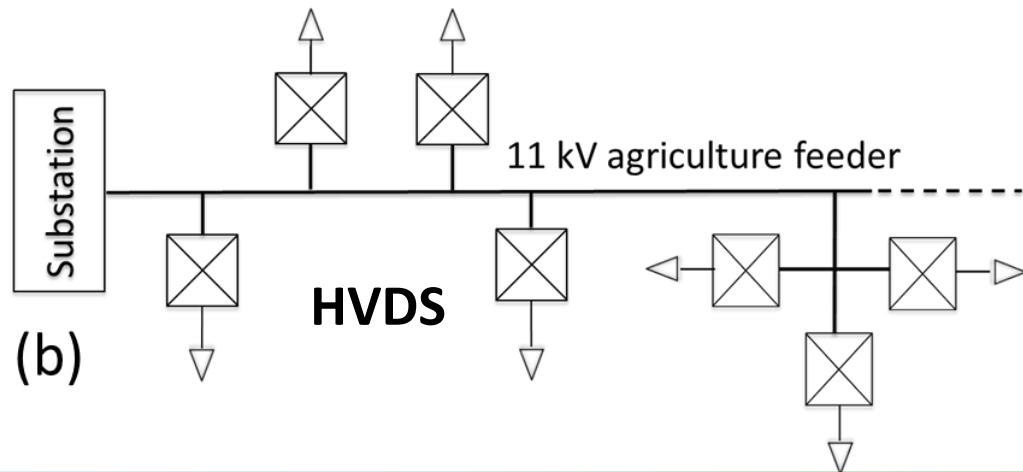
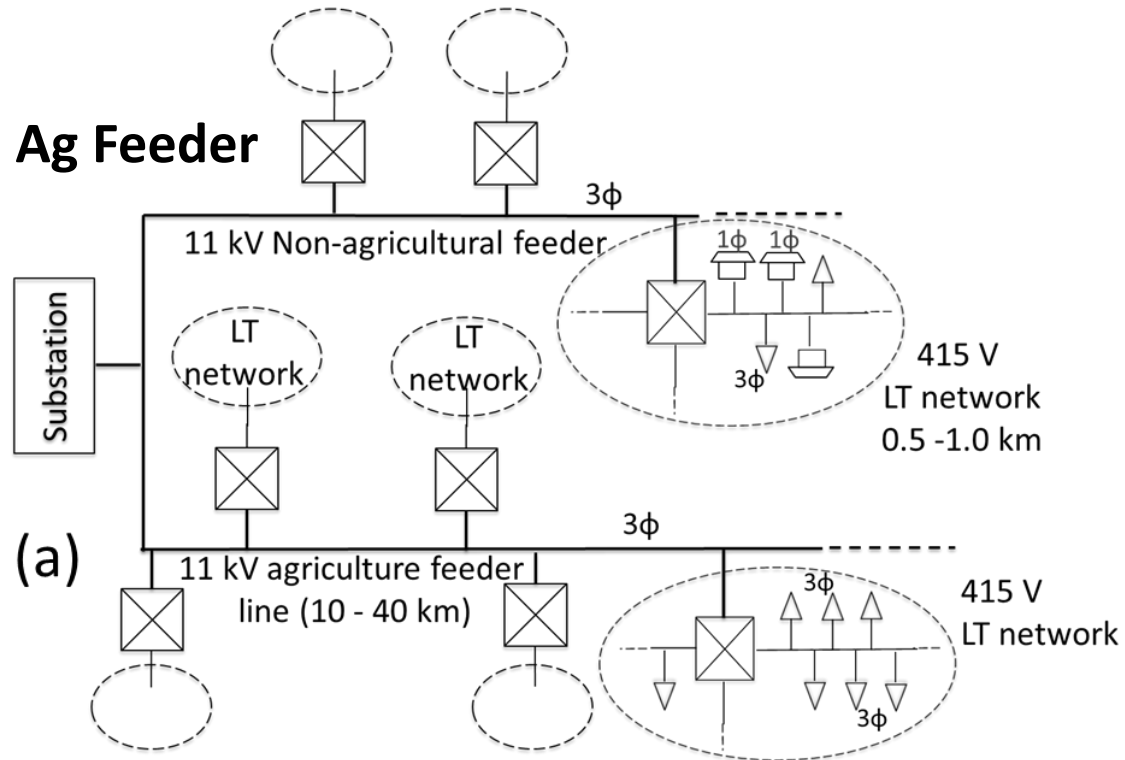
- Cost of any energy system: *Energy* + *Infrastructure*

For example:

- Offgrid solar PV: No energy cost, higher upfront infra cost
 - Ag feeders, HVDS– comparatively lower infra cost, but continuing energy cost
 - Losses: Lower on HVDS than Ag feeders, losses on Solar Ag feeders purportedly lower due to local generation
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- A proper comparison requires a consideration of **hours of irrigation in a year**, **distances between water sources** or **distance from grid**

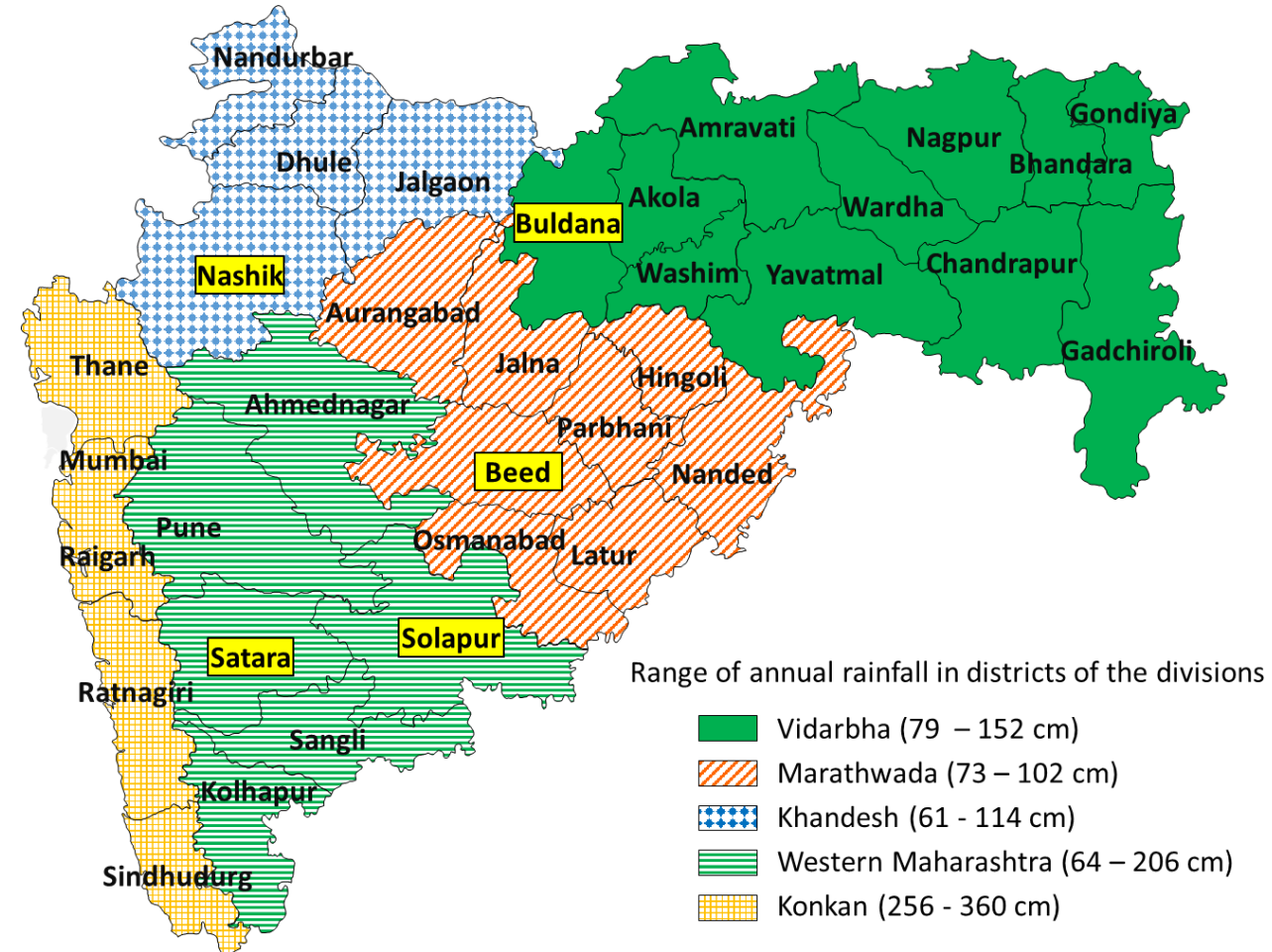
Who pays for what and how – whether MSEDCL, farmers, government – comes later, after a basic techno-economic analysis

Schematics of Systems

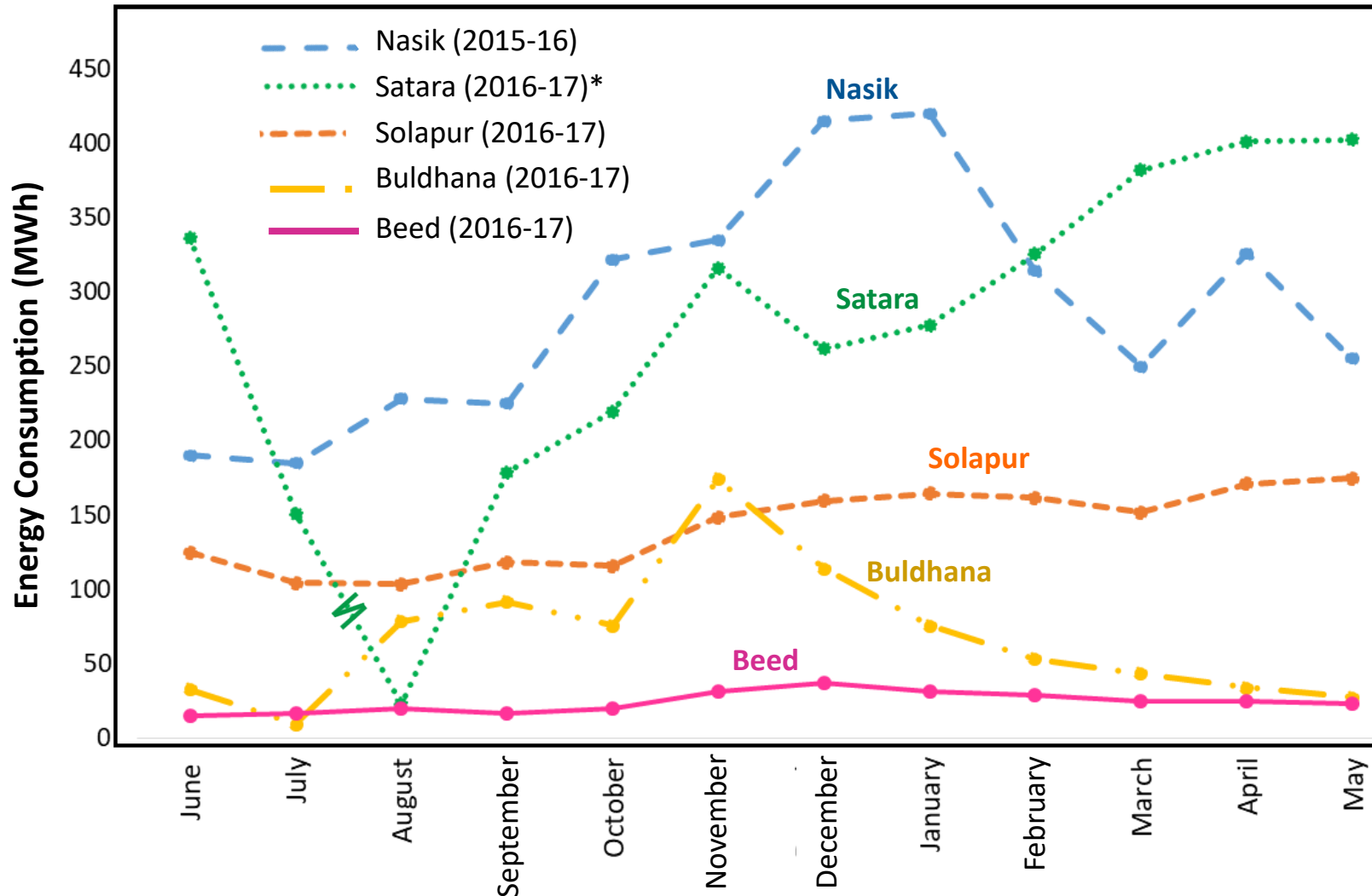


Different regions, different cropping / irrigation

- Energy consumption patterns differ based on the agro-climatic conditions
- Vidarbha and Marathwada: Peak irrigation in rabi , November to January
- Western Maharashtra: Sugarcane requires year round irrigation
- Nasik, Jalgaon: Heavily horticultural , year round irrigation
- Many variations within each region
- However, this is an illustration of how the cropping affects the feasibility of our technology choices

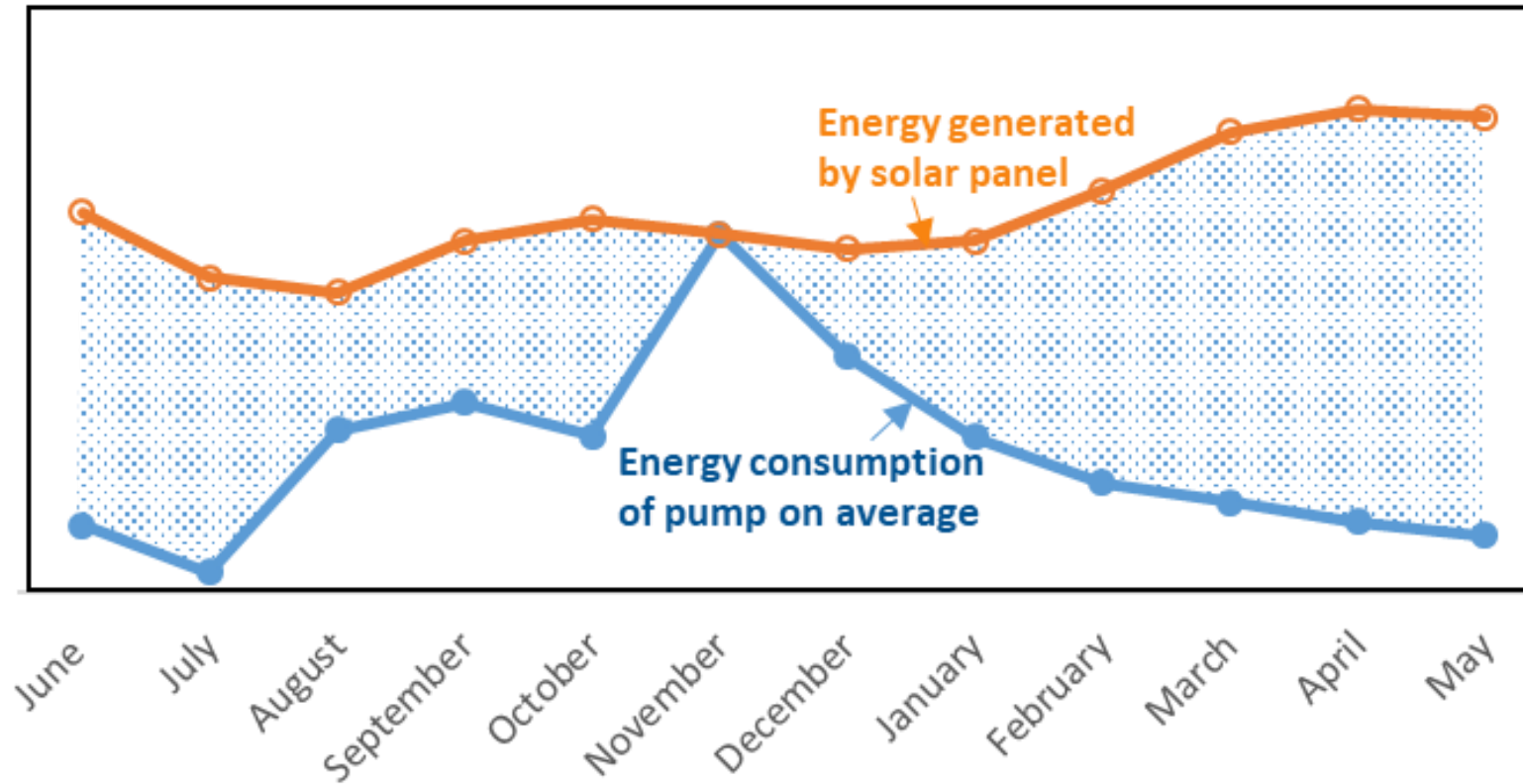


Energy consumption changes with agro-climatic conditions



- The year round energy consumption follows irrigation requirements
- The total energy consumption depends on the number of consumers etc.
- Buldhana is the most peaky

Solar PV energy generation and consumption



The sharpness of energy consumption is relevant for the following reasons:

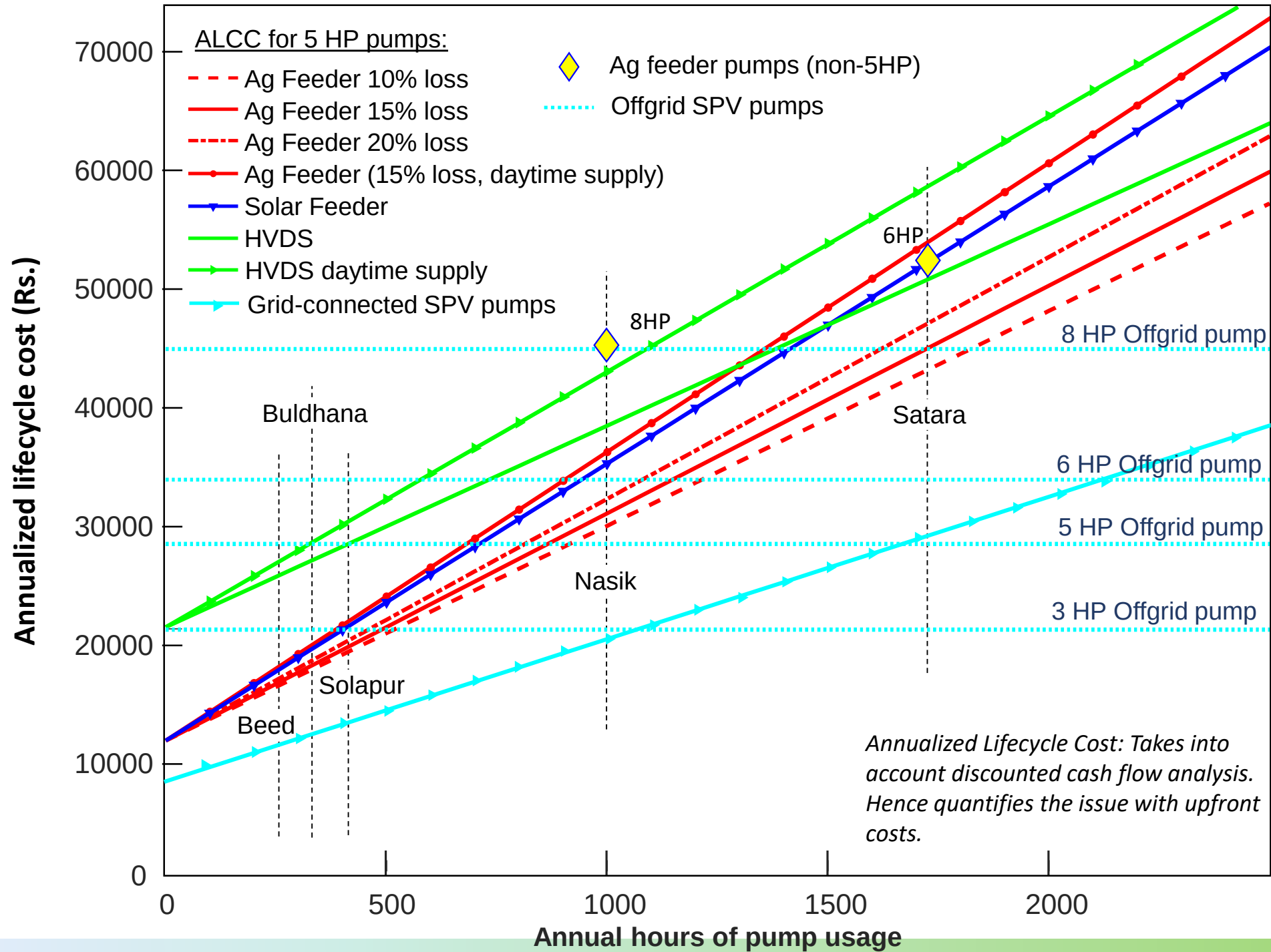
- Solar PV pumps are used for only part of the year and hence become more expensive compared to grid
- In case of Solar Ag feeders, most of the energy will be fed back to the grid

Costs and lifetimes of the various technologies

<u>Energy costs (Rs. / kWh) ^a</u>	
Energy cost in grid supply	4.04
Energy cost adjusted for Time of Day	3.18
Feed-in Tariff for Solar PV	2.74
Transmission charges	0.37
<u>Capital cost (5HP pump) (Rs. Lakhs):</u>	
Ag feeder	1
HVDS	2.5
Electric pump	0.2
Solar Off-grid pump (1HP-7.5HP)	1.2-4.9
Solar PV net-metered pumps	Offgrid + 0.5
Maintenance of solar offgrid pump	1%
Operational expenses for grid (Rs/year)	2842

<u>Losses:</u>	
Transmission loss	4%
Distribution loss - technical	10-20%
Distribution loss - technical in HVDS	4%
<u>Other assumptions:</u>	
Inflation	6.75%
Discount factor for Solar PV systems	12%
Average Daily Solar Resource (kWh)	7.15
Lifetime of solar PV systems (years)	20
Lifetime of Distribution Transformers	25
Lifetime of conductors	40
Lifetime of grid-connected pumps	10

Annualized Lifecycle Cost: Takes into account discounted cash flow analysis. Hence quantifies the issue with upfront costs



Costs per connection:

Infra cost: Rs.1 lakh

Annualized cost – Rs. 7000

HVDS cost: Rs.2.5 lakhs

Annualized: Rs.16,680

O&M: Rs.2842 / per year

Electric pump: Rs.20k

Annualized cost : Rs.1400

Solar PV Offgrid pumps:

Rs. 2.4–5 lakhs (3 – 7.5 HP)

Lifetime DT: 25 years

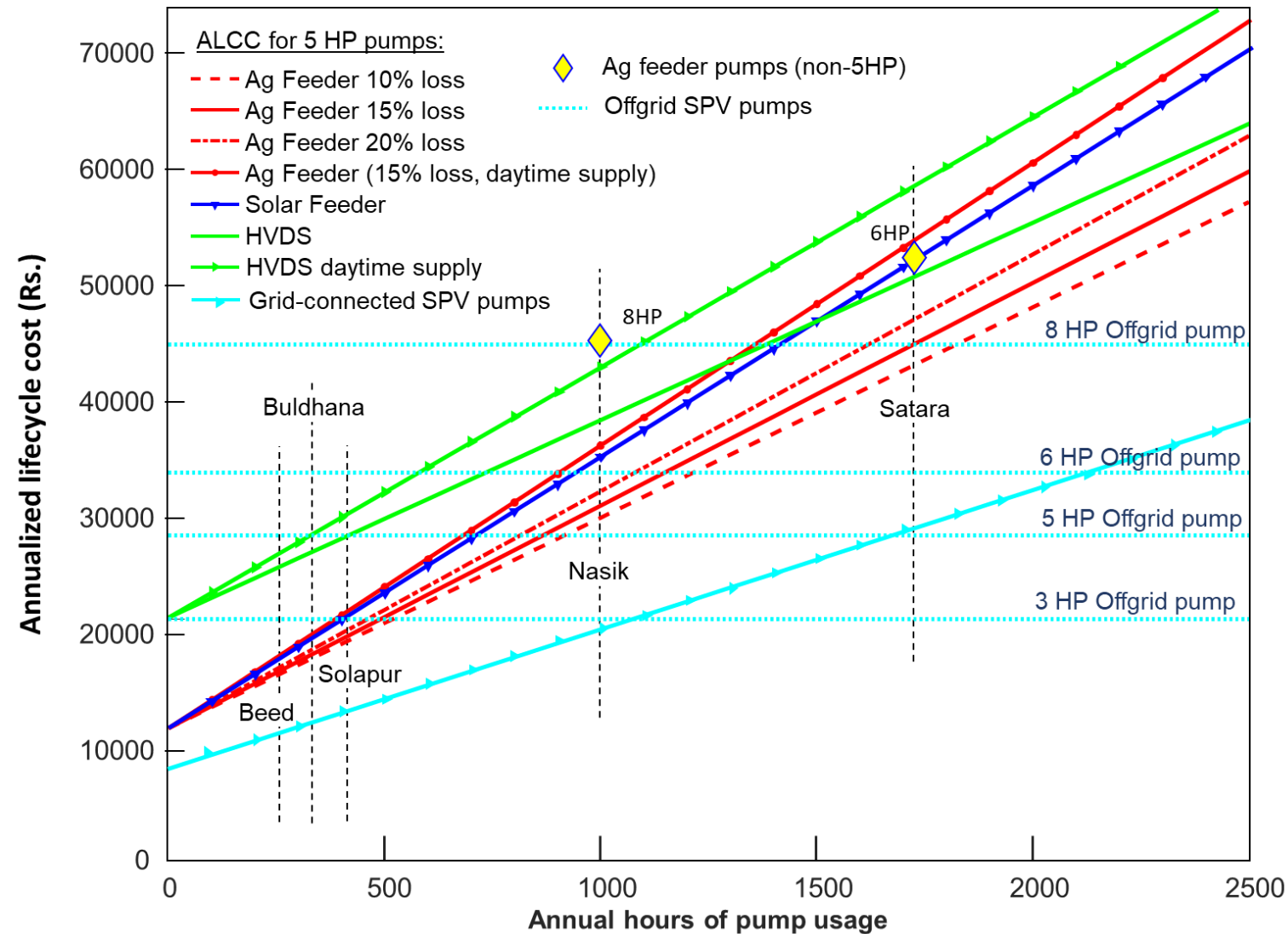
Conductors: 40 years

Pumps: 10 years

Solar PV pumps: 20 years

Discount factor = 12%

Inflation = 7%



- Solar net-metered pumps – the ALCC does not reflect true economic costs. Distributed renewable resources, like rooftop solar PV, are complicated.
- Ag Feeders is the most cost-effective
- Offgrid pumps become cost-effective at greater than 875 hours of pumping - but is constraining
- Solar Ag feeders – slightly lower than Ag feeders with daytime supply because we have not included transmission losses, however, complications remain
- HVDS is very expensive

Complications of Distributed Energy Resources

Solar Ag Feeders:

- ~3-8 MW solar PV plant per feeder
- Higher energy costs than Utility scale plants because 50 MW plant is a lot less costly than smaller plants due to economy of scale [Bids placed at Rs.3.30/kWh, over Rs.2.75/kWh]
- Also, the locational constraints means that developers cannot use the most cost-effective location
- Transmission losses may not really reduce because in the off-season, energy will not be used by farmers and hence the generated energy will use the transmission lines.
- Off-season may result in transmission bottlenecks too
- Will be more cost-effective to add utility scale PV plants and supplying daytime power to farmers

Complications of Distributed Energy Resources

Net-metered Solar PV pumps, akin to rooftop solar PV:

- The consumers that can be paid for the solar PV generated by their pumps/rooftop systems are in fact being subsidized by consumers who do not have such systems.
- This is because not all of the infra costs are included in the fixed cost, some of it is in the per unit charge, for e.g. transmission cost
- When consumers can both draw and supply power to the grid, the economics become very complex;
- Similar to Solar Ag feeders, in the off-season, distribution as well as transmission losses may increase because no (almost) energy is being used by the pumps. In case of many such systems in an area, this may result in high losses and transmission bottlenecks

HVDS

- Very expensive; 2.5 lakhs, versus 1 lakh for Ag feeder;
- Purported benefits: Good voltage, lesser DT breakdown, lower losses, theft through hooking reduced; Can all be achieved through Ag Feeders with better design at a lower cost
- Fewer pending connections will lead to less theft; Over-designing Ag feeder will cost less than HVDS and lead to all benefits
- Connecting 15 instead of 25 consumers on one DT, may increase the cost per connection by Rs.10 – 30k
- Technical losses do not make up for the high ALCC

Developing various strategies to improve supply at reduced infra cost with Ag feeders in PoCRA

Solar PV Offgrid pumps

- Very high subsidy, yet farmers reluctant to buy; High risk for several reasons;
- Safety of equipment in the field is an issue;
- The system is designed for a particular head, and leads to low flexibility
- If water source is fails and farmer digs new well, or farmer adds on drip/sprinkler at some time, then the system may not be sufficiently capable [In grid, changing the pump is not expensive]
- Using with pressurized systems (drip/sprinkler) with varying solar input means that energy is not optimally used



Ag Feeders

- Technical problems due to overloading: hooking, over-sizing of pumps, sometimes additional pump used by farmers
- Problems can go away by:
 - Ensuring farmers don't have to wait for years to get a connection
 - Over-designing the system a little (still cheaper than alternatives)
 - Introducing farmers to the benefits of capacitors and other load side strategies
- Other strategies to reduce infra costs like switching on the LT side



...Investigations ongoing in PoCRA

End of part (1), from paper

Investigations in PoCRA to improve supply

- Project on Climate Resilient Agriculture
- PoCRA in 15 districts of Vidarbha and Marathwada (2018-24).
- Dept. of Agri., Gov. of Maha. and World Bank
- Strengthening farmers through agricultural technology, water management, and development of water sources
- Component on energy for irrigation in IITB MoU with Dept. of Agri, Gov. of Maharashtra



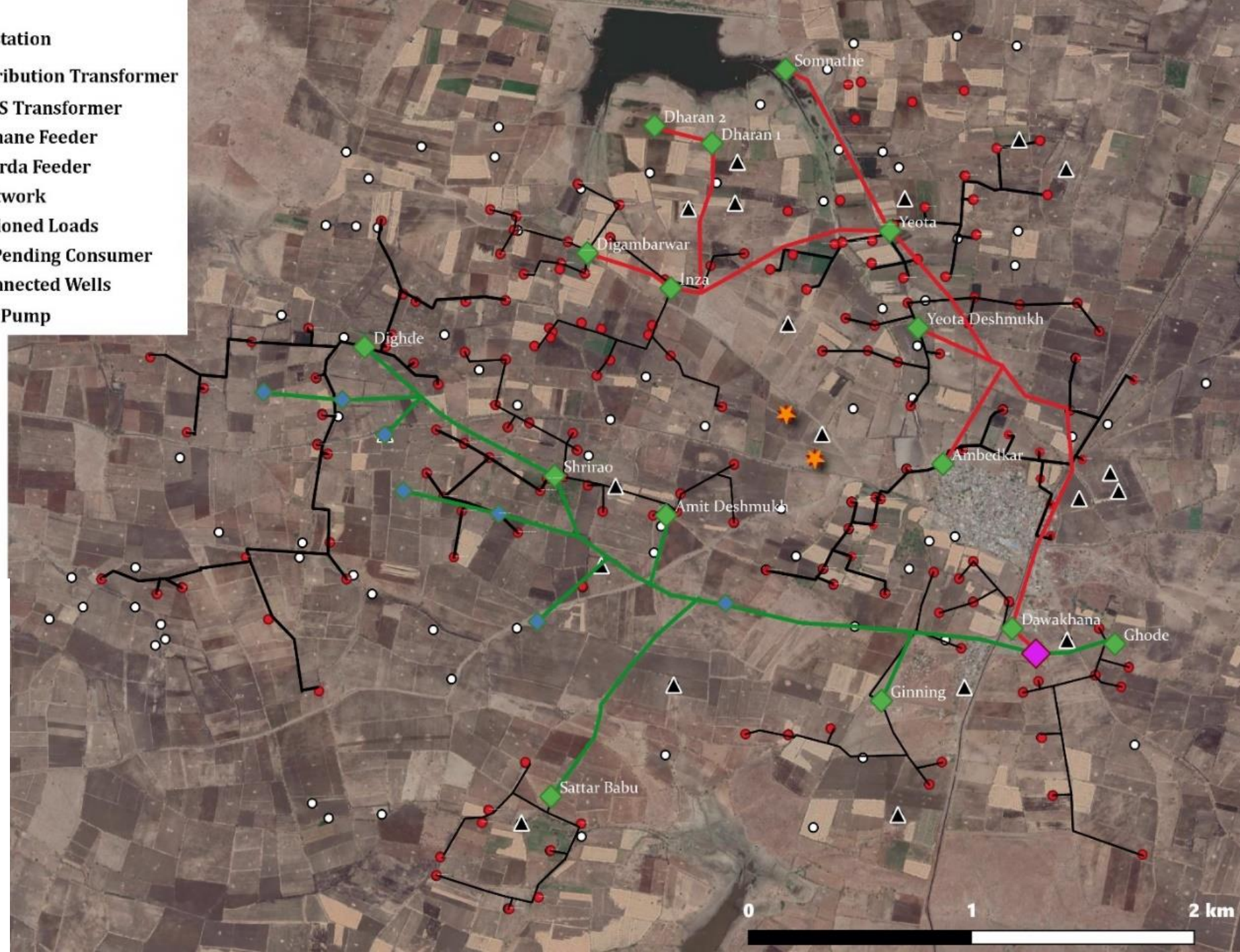
Following are results / learnings from investigations conducted as part of the work

Analysis conducted in Umbarda Bazar, Karanja, Washim

Distribution Infrastructure: Umbarda Bazar

Population: 4933
Cultivable area: 1300 ha
Farmers: 893
Connections: 189
Paid Pending: 19
Unconnected wells: 59

Umbarda substation
Feeders: Umbarda, Somthane
Dist. Transformers: 15
Avg. cost of connections
1.1 lakhs [81k – 1.9 lakh]
Cost Distribution
LT: DT: HT: substation
21%:48%:25%:7%



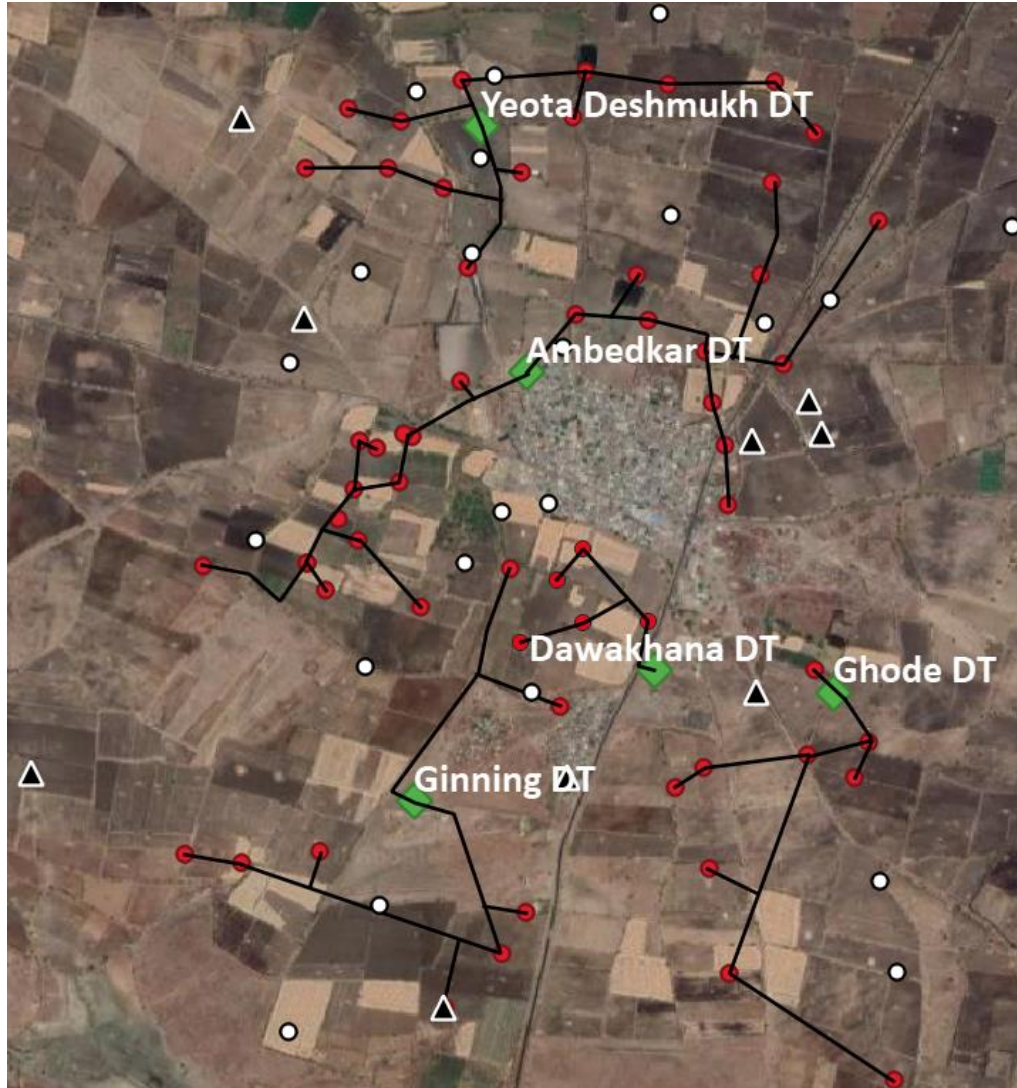
Strategy to add connections at minimal costs: Restructuring by moving LT lines

- Set of heuristics developed to redesign the LT networks for improved voltage profile and no overloading on all DTs
- Connections for all 19 Paid Pending + 59 Unconnected wells (78 new connections)
- Move some LT lines and add a few new ones; Add one new DT; Maximum 24 consumers per DT (5 HP load considered)
- Load flow analysis conducted to ensure proper voltage at maximal loading
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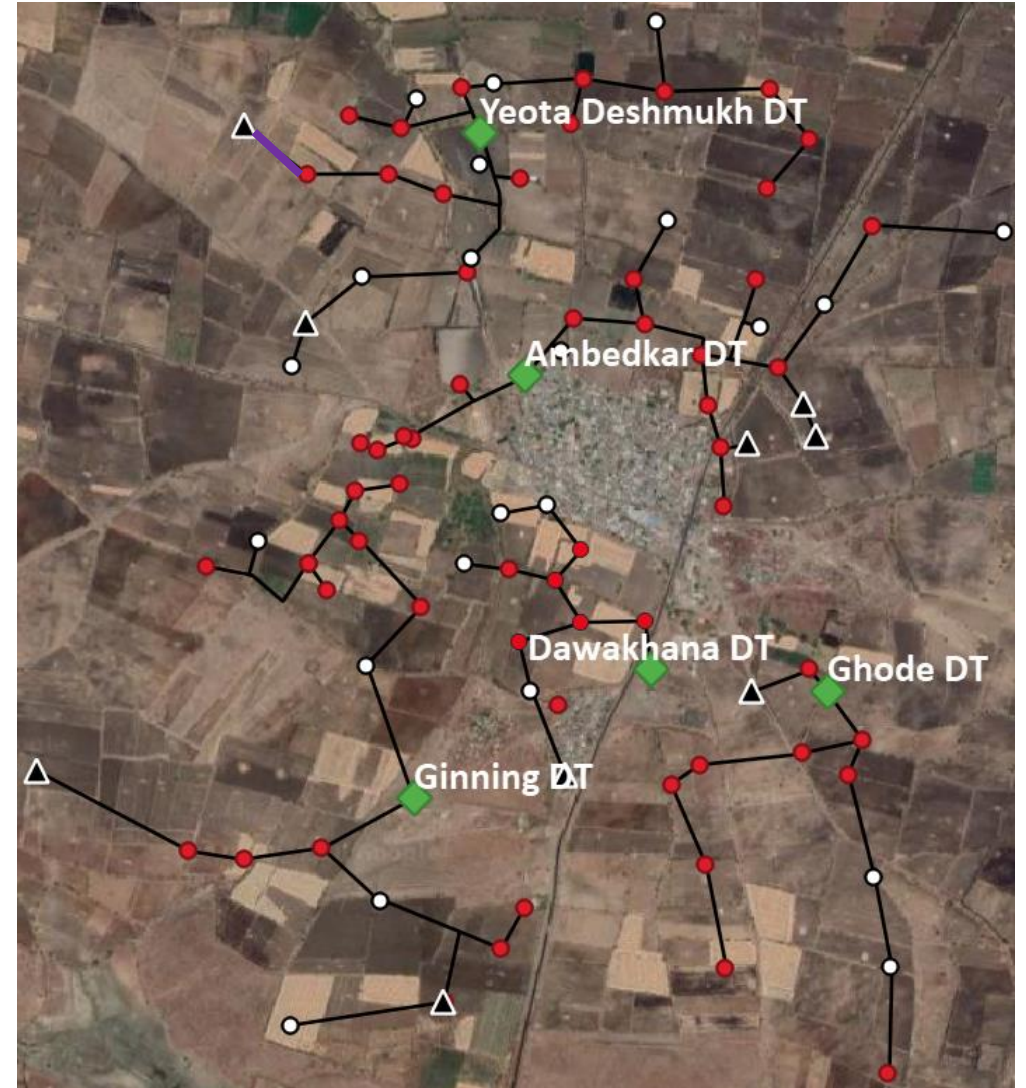
Restructured network on next slide

Detail of a section

Existing network



Proposed network



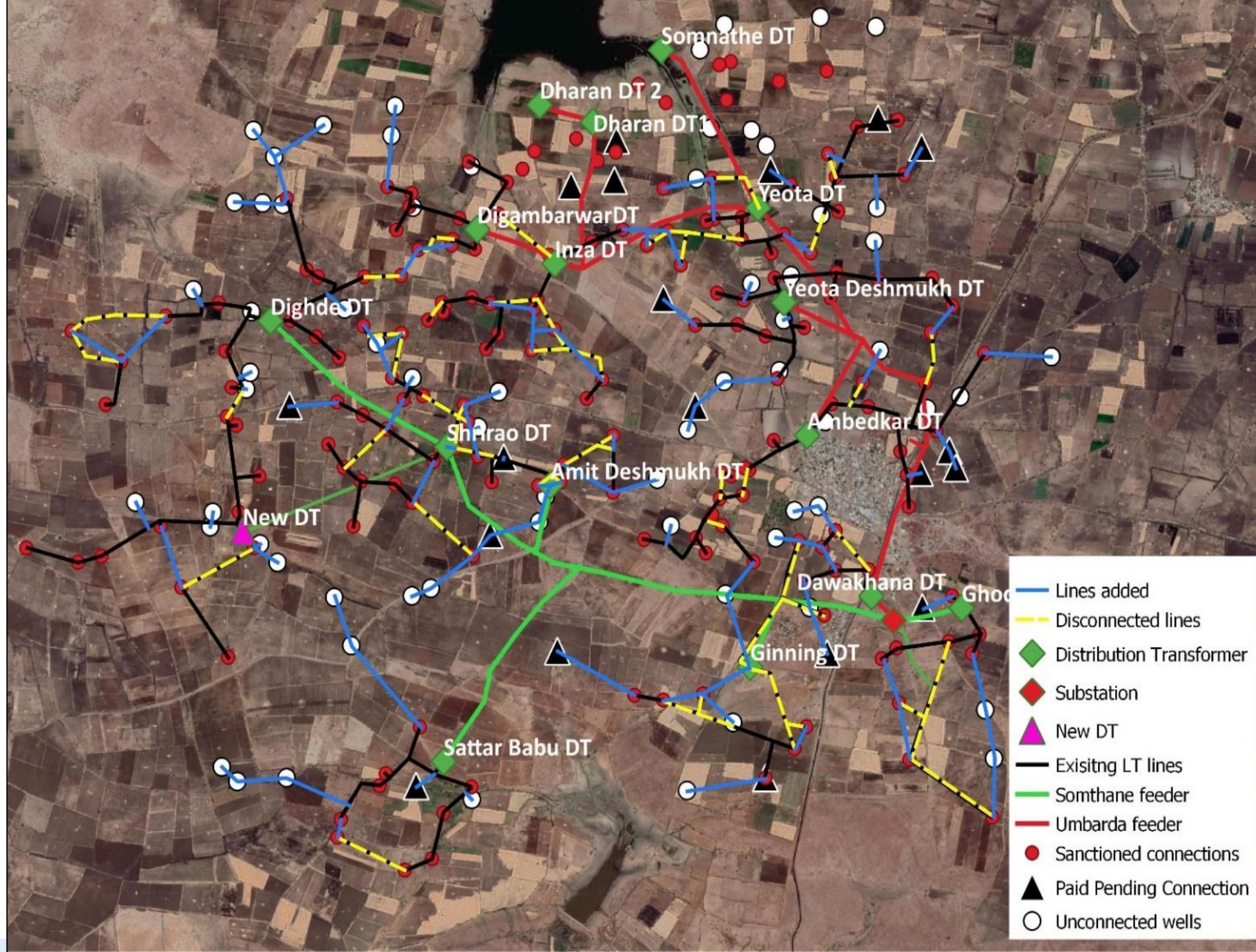
Restructured network of Umbarda Baazar

— Lines added
- - Disconnected lines

Rs. 27,500 per connection for 78 new connections + cost of moving LT lines

Compared to 1 lakh per new connection

Improved voltage and proper loading on all DT



Restructuring: A viable alternative

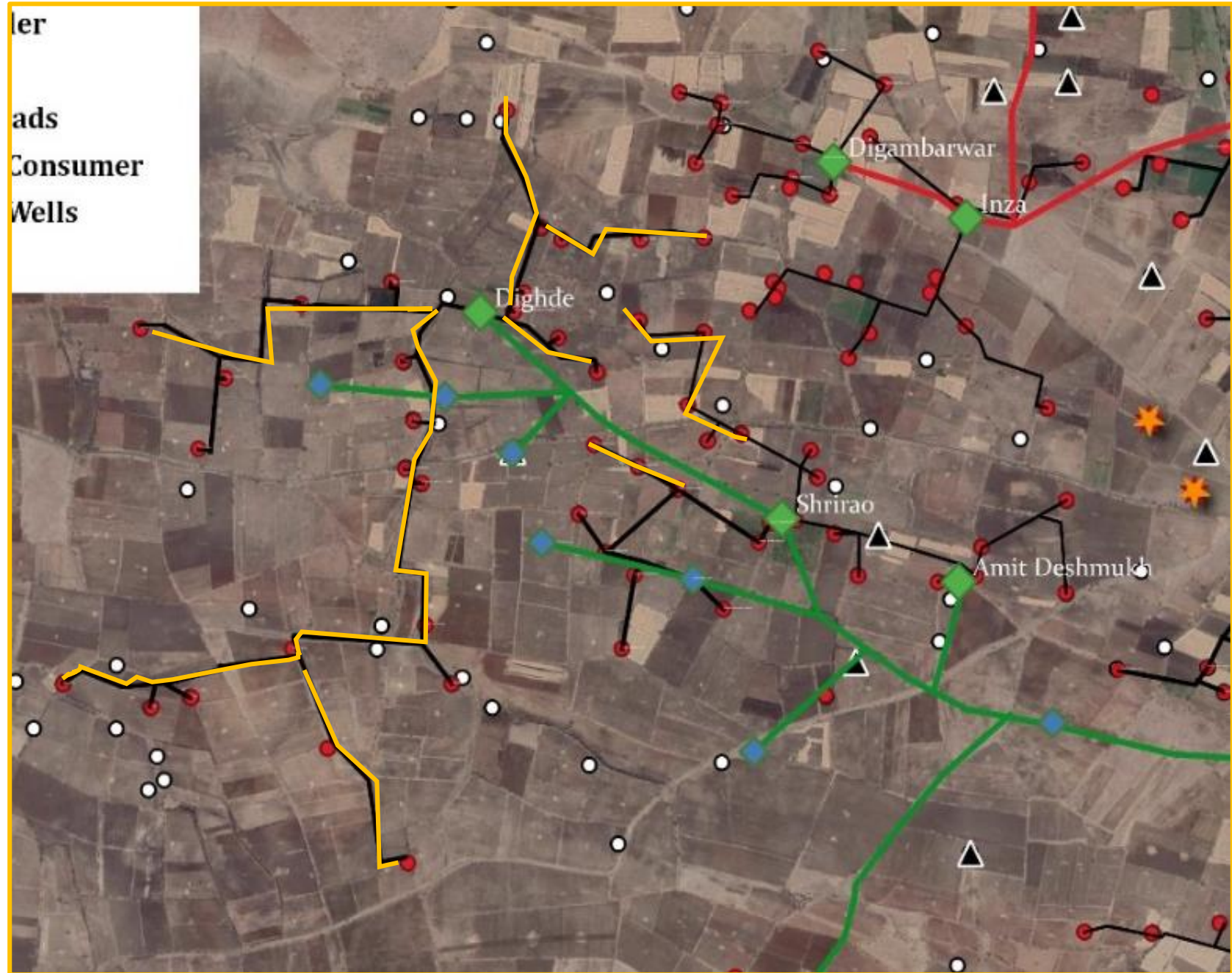
- Networks have developed over the years as the demand for connections grew, hence very long in places, some DT overloaded, some under-used
- Room for optimization of current network along with addition of new connections
- Consideration of load diversity to better estimate maximum number of consumers per DT
- Restructuring analysis initiated in additional villages:
 - Other villages on Umbarda feeder: Manbha, Yevta, Pimpri, Pilkheda, Wadgaon
 - Khanapur, Osmanabad
- In principle, moving LT lines possible – Feedback from MSEDCL engineers at Karanja, Washim, Yehala, Yavatmal, Khanapur, Osmanabad when shown the restructured network

Request to initiate pilot projects in these locations with the guidance and help of MSEDCL

Other possible optimizations under investigation

- Agricultural feeders have a peculiar quality of enforced power cut for 16 hours a day
- Infrastructure is *per force unused* for 1/3 of the time
- Switching on LT network to increase the infra usage factor can bring infra cost down to 1/3 of current costs
- A Distribution Transformer that currently has
 - Energy supplied for 8 hours a day to 1/3 of farmers on each DT through switches on LT network instead of at substation

- Keep HT feeder charged for 24 hours
- Switch out parts of the LT network
- Each farmer has 8 hours of supply per day
- Since the HT network and transformer are 80% of the cost, only additional 20% cost is required for the additional LT connections



Regional agro-climatic differences should reflect in the design

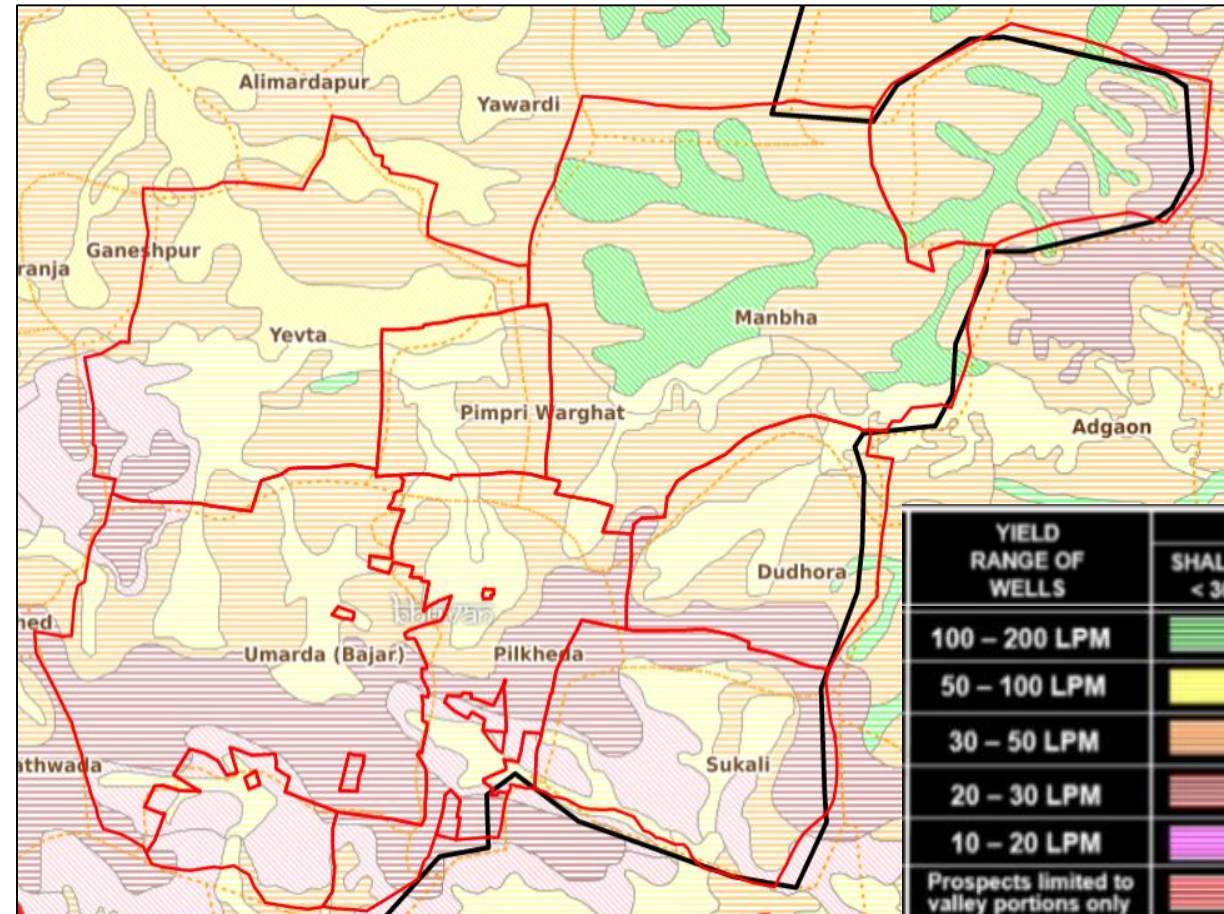
- Load Diversity is an important parameter
- MSEDCL design assumes no load diversity, 80 kVA connected on 100 kVA transformer
- Increases presumed cost per connection
- High load diversity in Sangli-Satara-Nasik (Horticultural areas) where year round crops
 - Farmers water regularly – every day 2 hours (drip for grapes), or every 15 days for sugarcane, hence they water at varying hours, allows high loading on a DT
- Low load diversity in Vidarbha-Marathwada where farmers use irrigation differently
 - In a dry spell in the kharif season, all farmers irrigate within a span of 3-4 days, hence high loading on DT

Involve regional engineering colleges in such analysis

Manbha village considered for grid analysis

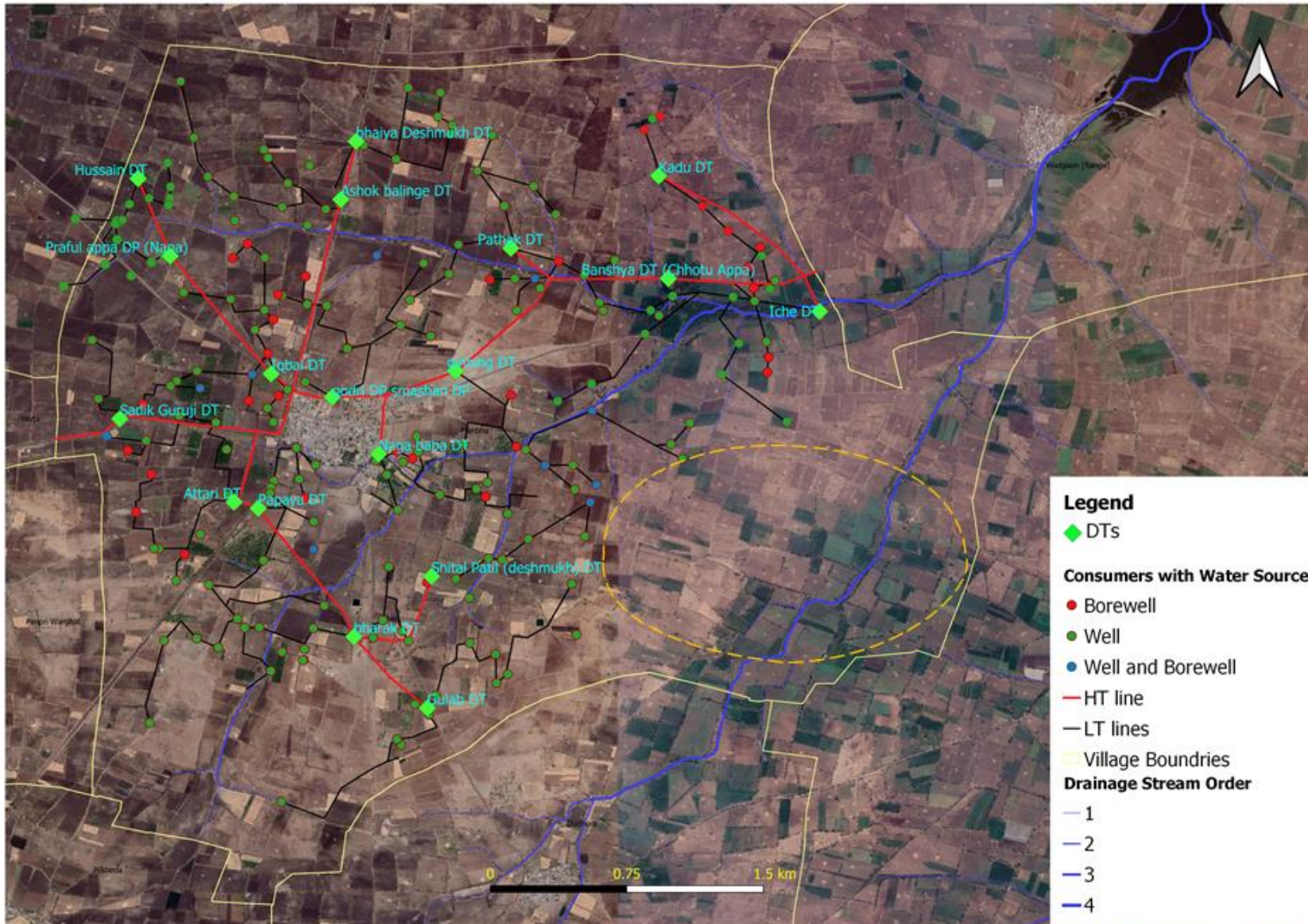
- Manbha and Umbarda, on Umbarda feeder
- Manbha has much higher percentage of irrigated land, horticulture, and greater number of wells
- Higher incidence of electricity issues

Village	Umbarda	Manbha
Cultivable area (hectares)	1299	1618
Horticulture (hectares)	2	100
Kharif (hectares)	1294	1502
Rabi (hectares)	71 (5%)	153 (9%)
Number of open wells (registered)	190	412
Number of borewells (registered)	24	56
Number of farmers accessing irrigation tank	43	-
Number of authorized connections	189	243



Horticulture: Sweet orange, lemon
 Rabi: Wheat, gram
 Kharif: Cotton, soybean, tur, udid, moong

Manbha village considered for grid analysis



- 18 DTs , 243 authorized connections
- ~160 unauthorised
- LT end does not get sufficient voltage to run sprinkler or drip after 1-2 hours after supply begins, as the load increases
- Farmers buy multiple locally manufactured low-cost pumps as pumps burnout frequently
- SouthEast corner of the village has no DTs, farmers here hook and drag long lines (upto 500m) for irrigation. Reason is yet to be investigated.
- Comparison with Umbarda in later slides

Work ahead:

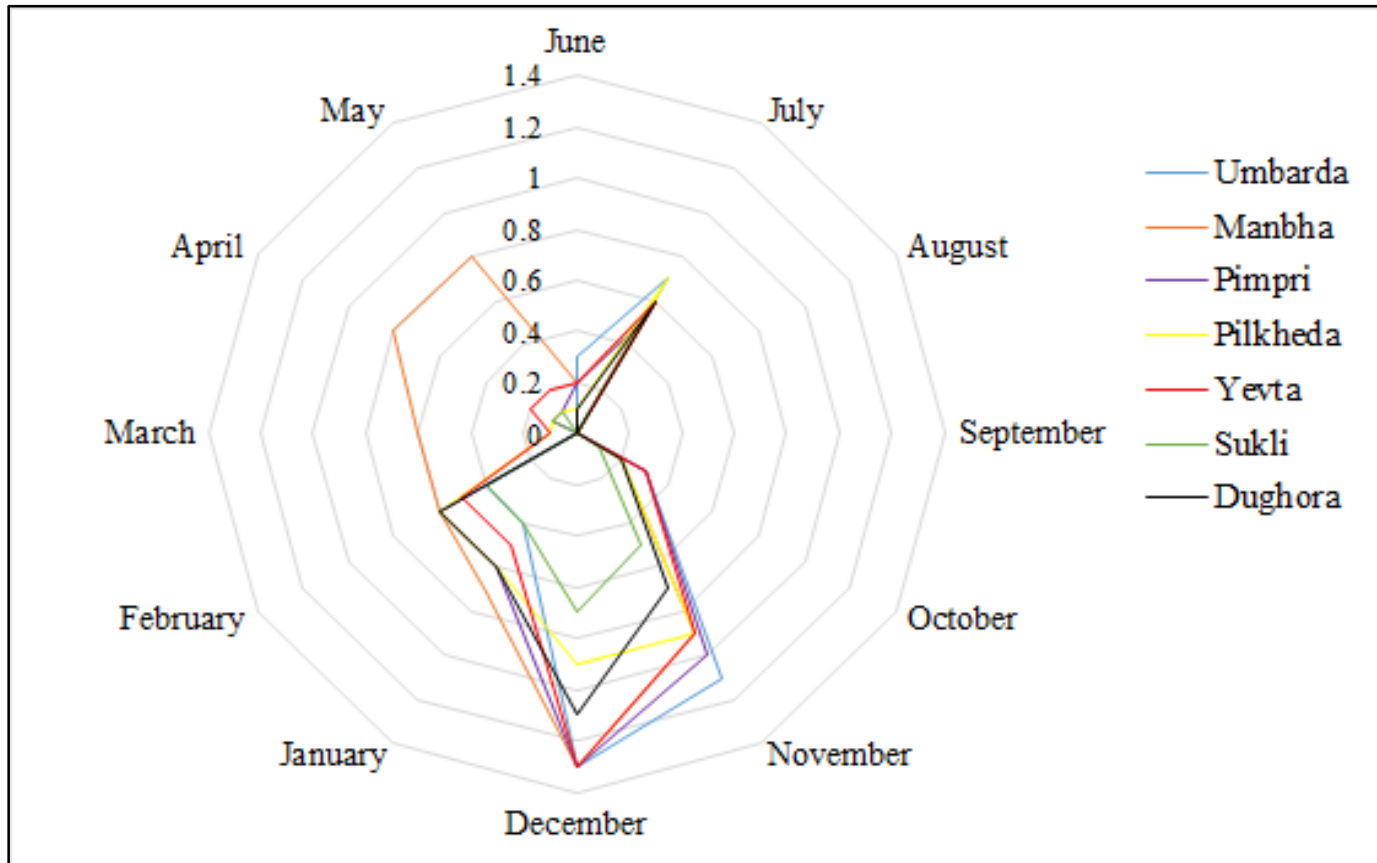
- Restructuring analysis
- Install meters on DTs for load diversity measurements if possible

Energy Indices : Sufficiency based on cropping

Seasonal Distribution Transformer Loading index

Capacity of DT considering the power drawl requirement for irrigating crops in that month

Accounted for: Cropping area, irrigation requirement, typical heads, sprinkler/drip/furrow, efficiency of pumps



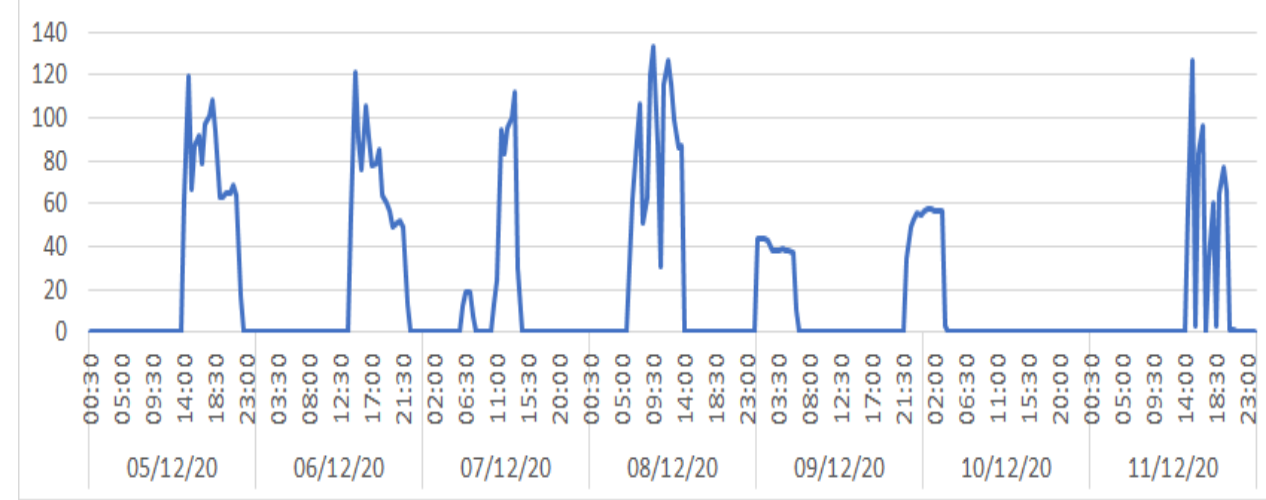
- Less than 1: DT capacity greater than required power drawl
- Greater than 1: Likely to be overloaded
- Assumption taken of 15 days irrigation, since farmers do prefer to irrigate in the daytime and there is no coordination among farmers in irrigation
- However better measurements for load diversity required
- In December most villages have heavy loading
- Results are averaged over a village

Total Load Estimation Tool

Cropping and irrigation for November 2020 in Karanja, Washim

DT Capacity (kVA)	100
Shift (hrs)	7
Power Factor	0.8
After Distribution Losses	75%
GW Head (m)	8
Pump efficiency (%)	10%

Day	Somthana AG	Manbha AG
Monday	14:00 – 22:00	22:00 – 06:00
Tuesday	14:00 – 22:00	22:00 – 06:00
Wednesday	06:00 – 14:00	14:00 – 22:00
Thursday	06:00 – 14:00	14:00 – 22:00
Friday	22:00 – 06:00	06:00 – 14:00
Saturday	22:00 – 06:00	06:00 – 14:00
Sunday	22:00 – 06:00	06:00 – 14:00



kVA loading for DT2

Cropping data on DT1

Sanctioned consumers: 17, Additnl: 12

Crop	Area (Ha.)	Applicati on Head (m)	Transfer Head (m)	Irrigations /Wk	Irrigati on (mm)	Volume (cu.m.)	Elec. Units Reqd.
Wheat	6	0	0	1	56	3360	747
Harbhara	4	30	0	0.8	45	1440	760
Cotton	10	0	0	0.5	50	2500	556
Tur	2	0	0	0.5	30	300	67
Orange	9	0	30	0.25	80	1800	1,900
Papaya	2	0	30	1	70	1400	1,478
Other	5	0	0	1	50	2500	556
Total area	38					Total	6,062
						Load	1.83

Cropping data on DT2

Sanctioned consumers:25, Additnl:4

Crop	Area (Ha.)	Application Head (m)	Transfer Head (m)	Irrigation s/Wk	Irrigation (mm)	Volume (cu.m.)	Elec. Units Reqd.
Wheat	10.2	0	0	1	56	5712	1,269
Harbhara	0.5	30	0	0.8	45	180	190
Cotton	14.6	0	0	0.5	50	3650	811
Tur	17	0	0	0.5	30	2550	567
Orange	4	0	30	0.25	80	800	844
Other	1	0	0	1	50	500	111
Total area	47.3					Total	3,793
						Load	1.21

Agricultural practice ↔ Electrical capacity

Improve performance on feeders at lower cost

Some interventions to improve performance on Ag feeders:

- Restructuring of LT lines to add new connections at minimal cost and improve the voltage profile through the entire network
- Further optimizations that can be done to reduce average infrastructure costs to MSEDCL and satisfy latent demand
- Planning at a higher level to integrate irrigation requirements with infrastructure



Capacitors and farmers – investigations done previously

- Many misunderstandings among farmers and lineman: Capacitors draw more energy, cause pumps to fail, means less water, stays charged and causes shocks
- Capacitors initiative in Solapur, Islampur, Nanded, Shegaon conducted by CTARA
- One feeder was selected at each location; Capacitors were installed on 75 – 80 farmers pumps;
- Installation from November 2017 – February 2018, Second set of surveys after a year;
- Mostly farmers were happy with the capacitors; In Solapur a DT that failed 3-4 times in a year, never failed, farmers were very satisfied
- However, 7 farmers' pumps failed in that year; Lineman told them it was because of the capacitors, they and 23 other farmers removed the capacitors

**Awareness building and promotion among farmers, linemen, electricians
can lead to a very valuable change (improved PF on Ag)**

Capacitors project work was done through student projects and seed grant funding from IIT

Unnat Maharashtra Abhiyaan and Unnat Bharat Abhiyaan provide a structure

राज्यातील अभियांत्रिकी महाविद्यालये तसेच
तंत्रनिकेतने व अन्य शैक्षणिक व संशोधन
संस्थामधील विद्यार्थ्यांना राज्याच्या
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महाराष्ट्र शासन
उच्च व तंत्र शिक्षण विभाग
शासन निर्णय क्रमांक : बैठक- ३६११/(५६/१५)/तांशि २
मंत्रालय विस्तार भवन, मुंबई ४०० ०३२.
दिनांक : १३ जानेवारी, २०१६



सत्यमेव जयते

अखिल भारतीय तकनीकी शिक्षा परिषद्
(भारत सरकार का एक सांविधिक निकाय)
(मानव संसाधन विकास मंत्रालय, भारत सरकार)
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प्रो. अनिल डी. सहस्रबुद्धे
अध्यक्ष

Prof. Anil D. Sahasrabudhe
Chairman

The Directors/Principals,
AICTE approved institutions.... (ALL).

Subject : Mass mobilization of engineering students from local engineering colleges and their involvement in campaign - for Jal Shakti Abhiyan Govt of India initiative.

- Government Resolution to involve students in engineering and other colleges in the development of the state...
- Local situations analysis
- Better learning for development as well as education
- GR No.: Gra.Pa.Dho.- 1114 / Case No.61 / Pa.Pu.-07 – Water Supply and Sanitation Department: Guidelines for third party technical inspection of Rural water supply schemes.
- AICTE promoted involvement of colleges for Jal Shakti Abhiyan

**Similar GRs should be introduced
in rural electrification including
Ag supply**

Past involvement of regional colleges in Ag supply projects / study

- Analysis of Infrastructure for Energized Irrigation (AIEI) workshop under UMA
 - Attended by 12 colleges from all parts of the state, 50 participants, faculty and students; January 2020
 - Following workshop did analysis of Ag feeder in the college area
- Study to examine Agricultural energy consumption, 2015-16
 - In collaboration with CTARA, IIT Bombay (15 colleges)
- Pilot project on farmer behaviour with regard to capacitors, Dec 2017 – Mar 2019
 - 4 colleges: RIT, Orchid Solapur, SGGIET Nanded, Mauli Shegaon – in collaboration with CTARA

Thank you